* A var args is internally array but unlike array we cannot use as a local variable i.e. var args we can use only for parameter variable.

Example :

```
void main(String ...args) //Var args  is valid here because using as  a parameter
{
   int ...arr = {10,20,30};  ////Var args  is Invalid here because using as  a parameter
}
```

* Whenver we use var args as a method OR constructor **parameter** then we have the flexibility to pass **direct values OR array variable to the var args because var args can accept both i.e direct values OR array variable.**

* Var args must be **only one and last parameter** otherwise we will get a compilation error.

WAP that describes var args can accept 0 to n of arguments :

```
VarArgs1.java
-------------
void main(String  ...args)
{
     accept();
     accept(1);
     accept(5,10);
     accept(5,10,15);
     accept(5,10,15,20);
     accept(5,10,15,20,25);
}

void accept(int  ...x)
{
     IO.println("Executed");
}
```

WAP to show var args can accept all different types of values.

```
VarArgs2.java
-------------
void main()
{
  acceptHetro(12, 23.90, 'A', true, "Java");
}

void acceptHetro(Object ...values)
{
  for(Object value : values)
     {
       IO.println(value);
     }
}
```

WAP to show var args can add parameter variable values.

```
VarArgsDemo3.java
-----------------
void main()
{
  addParameter(10,20);
  addParameter(100,200,300);
  addParameter(1000,2000,3000,4000);
}

void addParameter(int ...values)
{
     int sum = 0;

     for(int value : values)
     {
      sum = sum + value;
     }
     IO.println("Sum of the parameter is :"+sum);


}
```

WAP to show the real life use of var args for any shopping application to calculate the cart price.

```
VarArgs4.java
-------------
void main()
{
  addCartItemPrice(10,20);
  addCartItemPrice(100,200,300);
  addCartItemPrice(1000,2000,3000,4000);
}

public void addCartItemPrice(double ...prices)
{
 double billAmount = 0;
 for(double price : prices)
 {
    billAmount  = billAmount  + price;
 }

 IO.println("Dear customer, Your final bill is :"+billAmount);
}
```

WAP to show that var args can accept direct values OR array variable values.

```
VarArgs5.java
--------------
void main()
{
     IO.println("Passing Values directly...");
     accept(10,20,30,40,50);
     IO.println("Paasing Array Variable....");
     int []values = {60,70,80,90,100};
   accept(values);
}

void accept(int ...arr)
{
     for(int x : arr)
     {
          IO.print(x+" ");
     }
}
```

WAP to show that var args must be only one and last parameter.

```
VarArgs6.java
--------------
void main()
{
     accept(10,20,30,40,50);
}

/*
void accept(int ...x, int ...y)  //Invalid
{

}

void accept(int ...x, int y)  //Invalid
{

}
*/

void accept(int x, int ...y)
{
     IO.println("x value is :"+x);

     for(int z : y)
     {
          IO.println(z);
     }
}
```

Autoboxing and Unboxing :

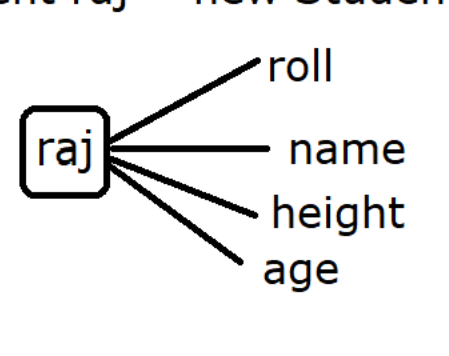
* Java is not a pure object oriented language because it is accepting primitive data types.

* Only objects are moving in the network but not the primitive data types.

Example :

```
class Student
{
   int roll;
   String name;
   double height;
   int age;
}

Student raj = new Student();
```

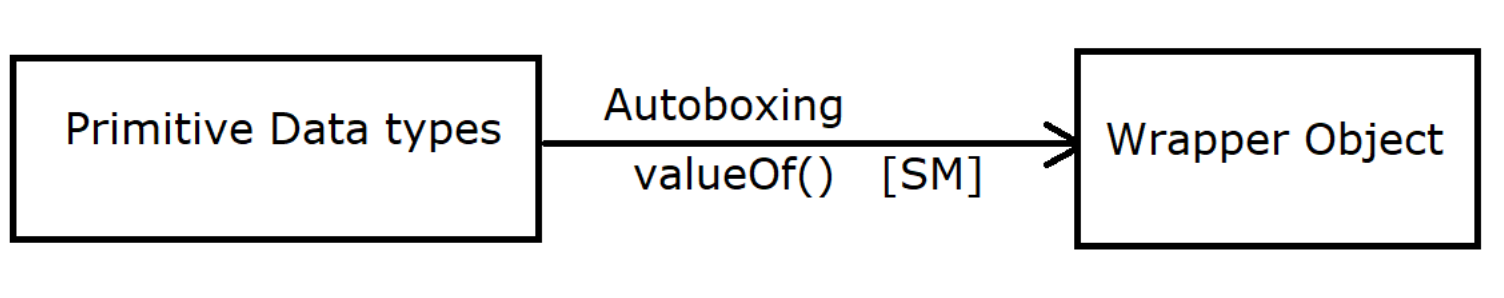


* In order to convert the primitive data types into corresponding object, Java has provided Wrapper classes concept.

* In order to convert from primitive to wrapper and vice versa, From JDK 1.5V onwards java has provided two concepts :
  1) Autoboxing
  2) Unboxing

Autoboxing :

* It is a technique through which we can convert the primitive data types into corresponding Wrapper object automatically.



* Till JDK 1.4V, A developer was responsible to convert the primitive data types into corresponding wrapper Object by writing valueOf() static method.

Example :

```
int x = 90;
Integer obj = Integer.valueOf(x);  //JDK 1.4V
IO.println(obj);
```

* JDK 1.5V onwards we can directly assign primitive data types into corespinding wrapper object.

Example :

```
int a = 100;
Integer b = a;   //JDK 1.5V [Autoboxing]
IO.println(b);
```

Here Internally, Java compiler will convert this primitive into Wrapper by writing the follwing code.

```
int a = 45;
Integer obj = a;
```

The Code generated by Java compiler

```
Integer obj = Integer.valueOf(a);
```

* The following diagram explains primtive to Wrapper conversion :

Primary Data type		Wrapper Object
byte	:	Byte
short	:	Short
int	:	Integer
long	:	Long
float	:	Float
double	:	Double
char	:	Character
boolean	:	Boolean

Methods :

1) public static Integer valueOf(int x) :

* It is a predefined static method of Integer class through which we can convert primitive int type into Integer Object.
  Note : All the Wrapper classes (8 Wrapper classes) have valueOf() method to convert the primitive into corresponding Wrapper.

2) public static Integer valueOf(String x) :

It is a predefined static method of Integer class through which we can convert any string type value into Integer Wrapper object.

```
String -> int        = parseInt()
String -> Integer   = valueOf();
```

WAP to show to convert primitive type into Wrapper :

```
Autoboxing1.java
--------------------
void main()
{
     int x = 90;
     Integer obj = Integer.valueOf(x);  //JDK 1.4V
    IO.println(obj);

     int a = 100;
     Integer b = a;   //JDK 1.5V [Autoboxing]
     IO.println(b);
}
```

WAP to convert the String into Wrapper type :

```
Autoboxing2.java
-------------------
void main()
{
     String str = "123";
     Integer obj = Integer.valueOf(str);
     IO.println(obj + 2);
}
```